VIDYAVARDHINI'S A. V. COLLEGE OF ARTS, K. M. COLLEGE OF COMMERCE & E. S. A. COLLEGE OF SCIENCE
UNIVERSITY OF MUMBAI · COMMUNITY ENGAGEMENT PROJECT (NEP 2020 GUIDELINES)

The hidden jungle in Mumbai's concrete heart

Local Biodiversity Mapping of Byculla Zoo & Surrounding Green Spaces
Urban Ecology, 42 Documented Wild Taxa, and the BioMap Web-GIS Platform

ACADEMIC PROJECT CREDENTIALS & VIVA PARTICULARS
Student Investigator: Riddhesh & Team | Course: T.Y. B.Sc. Computer Science (Semester V, NEP 2020)
Project Guide / Mentor: Prof. Shweta Iyer, Department of Computer Science | Vartak College, Vasai (W)
Live Web-GIS Platform: https://biomap-byculla.netlify.app | Academic Year: 2026–2027

What Mumbai measured — and what it ignored

Tracked

PM2.5, traffic noise, captive animal inventories, pruning schedules

Missed

nesting trees, Blue Mormon breeding sites, heritage Baobabs, active nests before pruning

Cost

pruning can destroy habitat; visitor attention skips living heritage; caretakers' ecological memory can disappear



Two students turned fieldwork into an open map

01

Field effort

30 hours across four zones

02

Tools

smartphone GNSS, 48 MP camera, Leaflet.js, OpenStreetMap, Python, GeoJSON

03

Protocol

fixed-radius point counts, linear belt transects, and three-tier verification

04

Principle

zero backend, zero cost, fully open



The map started with 84 local voices

malis, walkers, visitors, students, officials, guides, and investigators turned lived ecological knowledge into a shared evidence base.

Maruti Patil

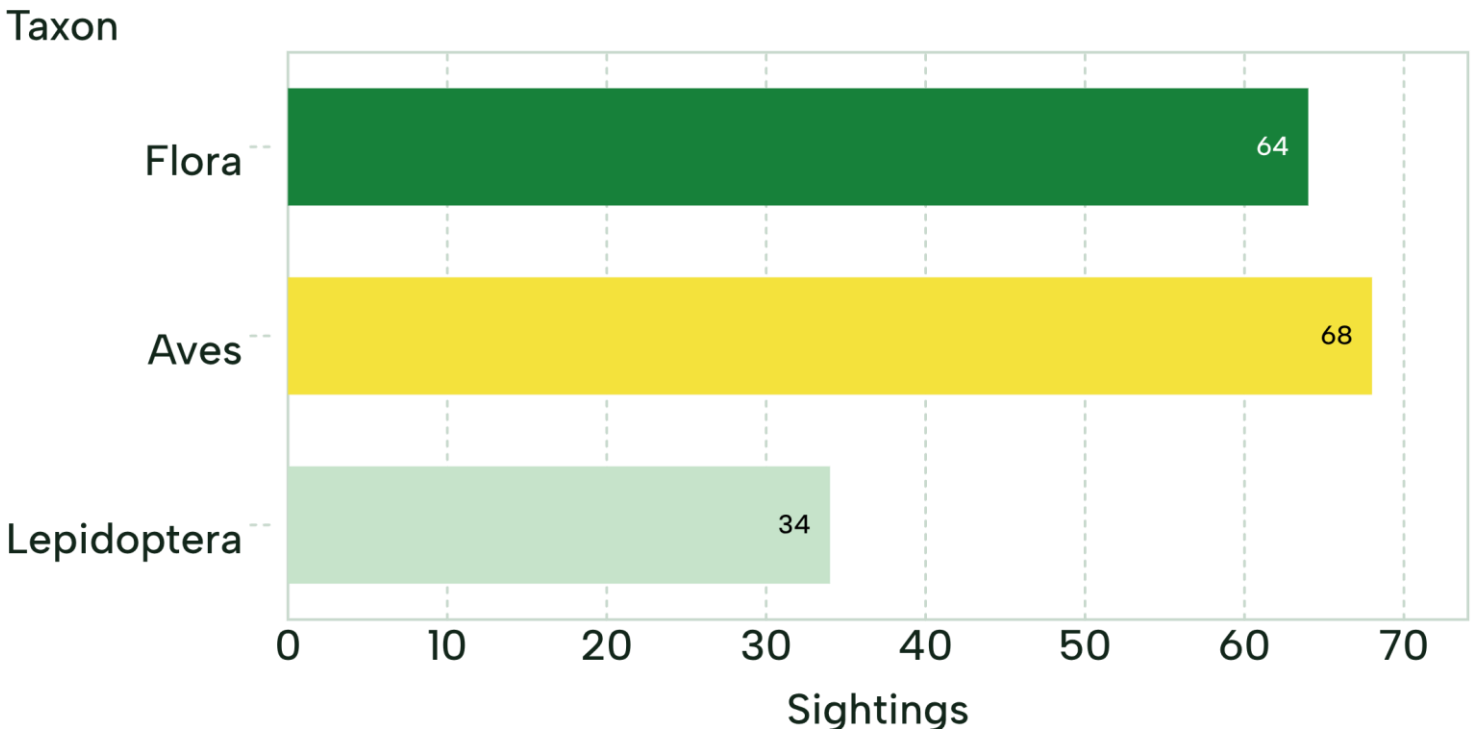
heritage trees and nesting sites need a map before pruning decisions are made

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Dr. Anita Rao

traffic and noise have pushed birds toward the botanical center

184 sightings revealed 42 species



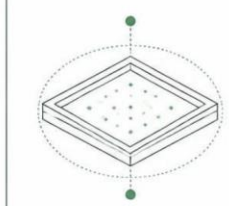
Named groups shown. Odonata plus herp/mammal records complete the census.



Flora · 18 species · H' = 2.71
Aves · 14 species · H' = 2.44
Lepidoptera · 7 species · H' = 1.82
Zone 1 holds 54% of all documented species.

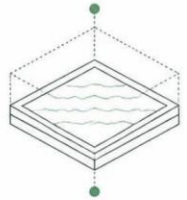


Life clusters where the canopy is thickest



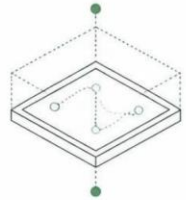
Zone 1 — Acoustic Oasis:

- H' = 2.68;
- 52 dB;
- Blue Mormon, Coppersmith Barbet, Baobabs, nesting hollows.



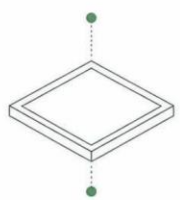
Zone 2 — Wetland Pulse:

- H' = 2.12;
- kingfishers, dragonflies, Painted Stork, floating macrophytes.



Zone 3 — Butterfly Corridor:

- H' = 1.94;
- Lantana, Ixora, mud-puddling Common Bluebottle.



Zone 4 — Concrete Wall:

- H' = 1.48;
- 78 dB;
- only House Crows and Common Mynas documented.

Noise and floodlights draw an ecological boundary

BOTANICAL CORE

52 dB

Zone 1's quiet canopy holds Blue Mormon, Coppersmith Barbet, Baobabs, and nesting hollows.

AMBEDKAR ROAD EDGE

78 dB

Particulate pollution, traffic noise, and night lighting create a barrier for sensitive species.

26 dB quieter

Canopy is not just scenery; it is habitat infrastructure.

BioMap turns observations into a civic tool

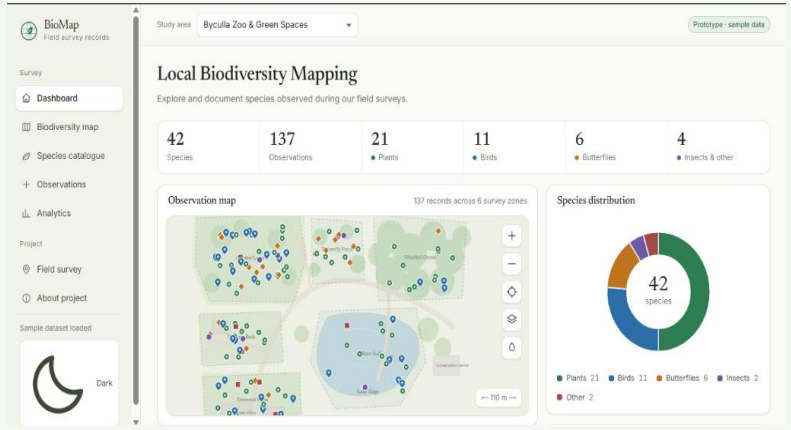
Explore: color-coded vector markers by flora, aves, and insecta

Filter: taxon and habitat zone in real time

Inspect: photos, GPS telemetry, and IUCN badges in responsive detail cards

Share: one-click GeoJSON and CSV export with no login or paywall

Include: WCAG 2.1 AA accessibility, high-contrast mode, and ARIA landmarks



LIVE WEB-GIS APPLICATION INTERFACE

- Interactive Leaflet.js engine with custom GPS pins
- Taxon filter: Flora, Aves, Lepidoptera, Reptilia
- Direct GeoJSON & CSV dataset export (zero paywall)
- Fully responsive & WCAG 2.1 AA accessible

biomap-byculla.netlify.app

Species become stories, not just records



Gharial



Sloth Bear



Humboldt Penguin



Painted Stork



Blue Mormon



Spotted Deer



Common Bluebottle

Recognizable field portraits give biodiversity data a face, a habitat, and a reason to care.

BIOMAP BYCULLA · LIVE PORTAL

Three tiers can move the map into practice

Now	QR placards on 25 heritage trees and a pre-monsoon nest check before pruning
Next	native larval host plants in Zone 3 and floating reed beds at the lake edge
Later	connect BioMap GeoJSON to the Mumbai Urban Biodiversity Index and add a citizen-science kiosk

THE HANDOFF

Tier 1 stops the bleeding.

Tier 2 heals the wounds.

Tier 3 builds the immune system.

PROJECT GUIDANCE & SUBMISSION

Project Guide / Mentor:
Prof. Shweta Iyer
Department of Computer Science

Vidyavardhini's Vartak College, Vasai (W)
University of Mumbai • NEP 2020 (Sem V)
Live Web-GIS: biomap-byculla.netlify.app

Thank You! • Questions & Viva Discussion